

Title

Leg Positioning Device Intervention Reduces Skin Breakdown and Wound Complications Following Cardiac Procedures

Abstract

Surgical complications including integumentary damage and contamination-related infections remain significant morbidity factors in patients undergoing open cardiac surgery. This trial compared a novel automated positioning device with standard passive dressing techniques in reducing these adverse events. Sixty cardiac surgery patients were randomized to either the intervention arm (receiving the positioning device) or control arm (standard dressing protocol). Primary evaluation focused on dermal injury development and infection rates measured at postoperative intervals. Secondary assessment included staff acceptability scores. The positioning device group demonstrated markedly lower rates of skin breakdown (2% versus 71%, $p = 0.002$, relative risk 3.18) compared to controls. Infection rates showed a trend favoring the device (3% versus 18%), though this did not reach statistical significance. Staff surveys indicated high satisfaction with the device approach (mean 4.82/5.0). Implementation of mechanical positioning strategies may substantially reduce integumentary complications in cardiac surgical populations.